Trent B. Thomas

Crosby Distinguished Postdoctoral Fellow, MIT EAPS
Dean's Postdoctoral Fellow, MIT School of Science
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Research Interests

Coupled evolution of volatiles & climate on rocky planets • Comparative planetology: Earth, Mars, exoplanets • Planetary atmospheres & habitability • Low-temperature geochemistry • Numerical modeling & Bayesian inference

Education

University of Washington — **Seattle, WA**
Ph.D., Earth and Space Sciences & Astrobiology (Dual-Title) — 2026

University of California, Los Angeles — **Los Angeles, CA**
B.S., Astrophysics, *Phi Beta Kappa* — 2020

Research Positions

Massachusetts Institute of Technology — **Cambridge, MA**
Crosby Distinguished Postdoctoral Fellow — 2026–present
School of Science Dean's Postdoctoral Fellow — 2026–present
Department of Earth, Atmospheric, and Planetary Sciences
- Hosted by O. Jagoutz and R. Pierrehumbert

University of Washington — **Seattle, WA**
NSF Graduate Research Fellow — 2020–2026
Department of Earth and Space Sciences
- Advised by D. Catling and V. Meadows

NASA Jet Propulsion Laboratory — **Pasadena, CA**
Research Assistant — 2018–2023
Exoplanet Discovery and Science Group
- Advised by R. Hu

Fellowships & Funding

Awarded
- Crosby Distinguished Postdoctoral Fellowship, MIT EAPS — 2026
- Dean's Postdoctoral Fellowship, MIT School of Science — 2026
- Winglee Endowed Graduate Fellowship, UW Earth and Space Sciences — 2023
- National Science Foundation Graduate Research Fellowship (NSF GRFP) — 2020

Pending
- Co-I, JWST Cycle 6 GO proposal, L 98-59 (PIs: M. Murphy & S. Ranjan). Submitted. — 2026
- Co-I, JWST Cycle 6 GO proposal, TRAPPIST-1 (PI: M. Gialluca). Submitted. — 2026
- Co-I, NASA HWO Precursor Science proposal, AGE-BioS (PI: I. Leifer). In preparation. — 2026

Publications

7 peer-reviewed publications (4 first-author)

- **Thomas, T. B.,** et al. *(Manuscript in preparation).*
  Retrieval of super-Io L 98-59 b's interior properties from atmospheric observations.

[7] Gialluca, M. T., V. S. Meadows, A. P. Lincowski, **T. B. Thomas,** P. C. Hinton, D. Brain, and D. Crisp (2026). Coupled Photochemical–Climate Modeling of Plausible Tenuous Outgassed Atmospheres on the TRAPPIST-1 Planets. *The Planetary Science Journal.* doi.org/10.3847/PSJ/ae6cef

[6] **Thomas, T. B.,** F. A. Macdonald, and D. C. Catling (2026).

Seafloor Weathering Can Explain the Disparate Durations of Snowball Glaciations. *Geology*. doi.org/10.1130/G53722.1

[5] **Thomas, T. B.,** V. S. Meadows, J. Krissansen-Totton, M. T. Gialluca, N. F. Wogan, and D. C. Catling (2025). Statistical Geochemical Constraints on Present-day Water Outgassing as a Source of Secondary Atmospheres on the TRAPPIST-1 Exoplanets. *The Planetary Science Journal*. doi.org/10.3847/PSJ/add261

[4] Adams, D., M. Scheucher, R. Hu, B. L. Ehlmann, **T. B. Thomas,** R. Wordsworth, E. Scheller, R. Lillis, K. Smith, H. Rauer, and Y. Yung (2025). Episodic Warm Climates on Early Mars Primed by Crustal Hydration. *Nature Geoscience*. doi.org/10.1038/s41561-024-01626-8

[3] **Thomas, T. B.,** and D. C. Catling (2024). Three-stage Formation of Cap Carbonates after Marinoan Snowball Glaciation Consistent with Depositional Timescales and Geochemistry. *Nature Communications*. doi.org/10.1038/s41467-024-51412-8

[2] **Thomas, T. B.,** R. Hu, and D. Y. Lo (2023). Constraints on the Size and Composition of the Ancient Martian Atmosphere from Coupled CO_2–N_2–Ar Isotopic Evolution Models. *The Planetary Science Journal*. doi.org/10.3847/PSJ/acb924

[1] Hu, R., and **T. B. Thomas** (2022). A Nitrogen-Rich Atmosphere on Ancient Mars Consistent with Isotopic Evolution Models. *Nature Geoscience*. doi.org/10.1038/s41561-021-00886-y

Presentations

Invited

- Virtual Seminar for Precambrian Geology (VSPG) 2026
- Earth and Ocean Sciences Department Seminar, University of Victoria 2025
- Center for Geoastronomy Astrophysics Colloquium, LMU Munich 2025
- National Oceanography Center Department Seminar, University of Southampton 2025
- Earth Sciences Department Seminar, Dartmouth College 2025
- Astrobiology Program Seminar, University of Washington 2024
- Virtual Planetary Laboratory Seminar, NASA/UW 2023
- High Performance Computing Seminar, NASA JPL 2022
- Mars Atmosphere Seminar, Caltech 2020

Contributed (all first-author)

- Astrobiology Science Conference, Madison, WI (poster) 2026
- AGU Fall Meeting, New Orleans, LA (poster) 2025
- STScI Atmospheric Escape and Replenishment Workshop, Baltimore, MD (talk) 2025
- Life and Planet Conference, London, England (poster) 2025
- European Geosciences Union General Assembly, Vienna, Austria (talk) 2025
- Astrobiology Science Conference, Providence, RI (poster; finalist) 2024
- Extreme Solar Systems V, Christchurch, New Zealand (poster) 2024
- Goldschmidt Conference, Lyon, France (talk) 2023
- 54th Division for Planetary Sciences Meeting, London, Ontario, Canada (talk) 2022
- Astrobiology Science Conference, Atlanta, GA (talk) 2022
- 53rd Lunar and Planetary Science Conference, The Woodlands, TX (talk) 2022
- AGU Fall Meeting, Virtual (talk) 2020
- 52nd Division for Planetary Sciences Meeting, Virtual (talk) 2020
- UCLA Undergraduate Research Week, Virtual (talk) 2020
- 9th International Conference on Mars, Pasadena, CA (poster) 2019

Awards & Honors

- David A. Johnston Award for Research Excellence, UW Earth and Space Sciences 2025
- Student Poster Competition Finalist, Astrobiology Science Conference 2024

- Dean's Prize for Excellence in Undergraduate Research, UCLA 2020
- Early Career Collaboration Award, NASA Astrobiology 2019
- Rudnick-Abelmann Scholarship, UCLA Physics & Astronomy Department 2019

Teaching

University of Washington **Seattle, WA**

- Teaching Assistant 2024
  ESS 101 Introduction to Geology and Societal Impacts (instructor: T. Swanson)
  Taught two weekly sections of 20–30 students, held weekly office hours.
- Course Creator & Teaching Assistant 2023
  ESS 103 Earth's Origin and Transformation over 4.6 Billion Years (instructor: M. Richards)
  Created ten 80-minute lectures, syllabus, and other course material.
  Performed all teaching assistant duties. Guest lectured "The history of life on Earth."

Coyote Central Youth Arts Organization **Seattle, WA**

- Course Creator & Instructor Summer 2024, 2025, 2026
  Generative Design: Creating Art with Code
  Created and taught a 20-hour course in Python for K–12 students with no prior coding experience.
  See final projects: 2024, 2025.

Mentorship

- Cem Ozyalcin, graduate student, MIT EAPS 2026–present
  Research project: extending Snowball Earth cap carbonate modeling
- Veronica Fula, undergraduate, University of Washington 2025
  Research project: deep learning segmentation of alluvial fans on Mars (GitHub)
- Jasmine Singh, undergraduate, Purdue University 2023
  Graduate school application mentoring (GEMS program); now graduate student at UT San Antonio
- Graduate peer mentor, UW Earth and Space Sciences 2021–2026

Service

- Peer reviewer for *Science Advances, Nature Communications, Earth and Planetary Science Letters, Icarus,* and *The Open Journal of Astrophysics*
- Early Career Investigator Committee, NASA LIFE Research Coordination Network 2025–present
- Expert Screener, CDRXIV, Preprints and Data for Carbon Dioxide Removal 2025–present
- Primary Convener, "Global Environmental Changes and Increased Biological Complexity in the Neoproterozoic and Paleozoic," AbSciCon & AGU 2024, 2025
- Department Representative, UW College of the Environment Student Advisory Council 2024–2025
- Student Representative, UW Earth and Space Sciences Faculty Search Committee 2024
- Contributor, NASA NExSS & NASA NFoLD Science Communication 2022–2023
- Committee Member, UW Earth and Space Sciences 2020–2026
  Awards, computing, graduate-nominated speaker (×2), retreat, peer mentor

Community Outreach

- Speaker, Everett Rock and Gem Club, Everett, WA 2025
- Science Guest, Mad Science Improv, Bandit Theater, Seattle, WA 2025
- Speaker ("Mars: Why the Hype?"), Astronomy on Tap, Seattle, WA 2022
- Volunteer Teacher (8 hours, 120 students), Nelson Middle School, Seattle, WA 2022
- Page Creator (Prebiotic atmosphere), Wikipedia 2022
- Speaker, Delran School System Family STEM Night, Delran, NJ 2022
- Social Media Manager, UW Astrobiology 2021–2022
- Creator & Moderator, UW Astrobiology Public Science Panel Series 2021
- Volunteer Guide, UCLA Planetarium 2019–2020

- Volunteer Scientist, UCLA Exploring Your Universe 2019

Media Coverage

- *GSA News*, R. Molinek: Understanding "Snowball Earth" Extreme Climates 2026
- *Geoscientist*, H. Bird: Seafloor clues to Snowball Earth 2026
- *UW News*, H. Hickey: Explaining dramatic planetwide changes after world's last 'Snowball Earth' event 2024
- *NASA Astrobiology*, A. Gronstal: The Size and Shape of Mars' Ancient Atmosphere 2023
- *LPI Planetary News*: Isotopic Evidence that Ancient Mars' Atmosphere was More Earth-Like 2022